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C PROGRAMMING PROJECT PRESENTATION

Project Title: mini library catalogue

Course: C Programming Lab / Mini Project

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Department: Computer Science

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INTRODUCTION

* The mini library catalogue is a C-based console application designed to manage a small collection of books efficiently

* It supportsadding,displaying,searching,updating,anddeletingbook records.

* Using filehandling, booksdata isstoredpermanently tpsimulateasmall-scale digital library system

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OBJECTIVE

* To understand and implement file handling in c

* To design and develop a library management system using structures.

* To apply modular and structured programming concepts.

* To maintain book records persistently and efficiently.

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PROBLEM STATEMENT

* Managing a library manually becomes difficult as the number of books increases.

* Traditional record-keeping methods may cause dataloss or errors.

* This project provides a simple and reliable console-based system to manage book records effectively using file storage.

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SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- OS: Windows / Linux
- Compiler: GCC / Turbo C / Code::Blocks / Dev C++

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SYSTEM DESIGN

The system follows a modular design.

Modules:

- Input Module–Takes user details for books.
- Processing Module–Performs operations(add, search, delete, etc.)
- File Handling Module–Stores and retrieves records from files.
- Output Module–Displays data on the console

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MODULE DESCRIPTION

1. Add Book–Adds a new book record.
2. Display Books–Shows all stored records.
3. Search Book–Finds a book by ID.
4. Delete Book–Removes a book entry.
5. Update Book–Edits book details.
6. Count Books–Displays total number of books.
7. List by Author–Shows books written by a specific author.
8. List by Year–Lists books published after a given year.

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ALGORITHM / FLOWCHART

Algorithm steps:

1. Start
2. Display Menu
3. Take user choice
4. Perform respective operation
5. Repeat until user exits
6. Stop

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IMPLEMENTATION

Techniques Used:

- Structures for storing book data
- File handling(fopen, fwrite, fread, etc.)
- Modular programming using functions
- Menu-driven user interface

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OUTPUT SCREENS

```
======== MINI LIBRARY CATALOGUE =========
 1. Add a New Book
 2. Display All Books
 3. Search Book by ID
 4. Delete Book by ID
 5. Update Book Details
 6. Count Total Books
 7. List Books by Author
 8. List Books Published After Year
 9. Exit
==========================================
Enter your choice: 1

Enter book ID: 101
Enter book title: C Programming
Enter author name: Dennis Ritchie
Enter publication year: 1978
```

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RESULTS & DISCUSSION

- The program successfully performs all file-based operations.
- Data remains stored permanently even after program termination.
- The system is user-friendly and efficient for managing small libraries.
- Challenges included handling file updates and avoiding data duplication

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

- The project achieved its intended objectives.
- It provided a deep understanding of file handling and data management in C.

Future Enhancements:

- Add a login system for librarians.
- Include categories/genres.
- Generate reports in CSV or text format.
- Create GUI or webinterface.

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REFERENCES / ACKNOWLEDGEMENT

References:

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- C Programming–E. Balagurusamy
- Online Tutorials and Documentation

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THANK YOU